

Hypothesis-Search Report

Myosin inhibitors act chiefly by relieving outflow obstruction rather than reversing the sarcomere substrate

Disease: Hypertrophic Cardiomyopathy (Complex) **Hypothesis ID:** myosin_inhibitor_obstruction_vs_substrate **KB status (input):** EMERGING **Report date:** 2026-10-09 **Run mode:** Literature-hypothesis search (no data files provided; no omics/patient-level dataset or computational analysis performed — see *Data and Tool Use*).

1. Executive Judgment

Verdict: PARTIALLY SUPPORTED (lean-supported for the symptomatic/functional endpoint; unresolved for the substrate-reversal claim).

The strong form of the hypothesis — that the *clinical benefit* of cardiac myosin inhibitors (CMIs; mavacamten, aficamten) in obstructive HCM (oHCM) is **predominantly load-dependent relief of LV outflow tract (LVOT) obstruction** — is well supported for symptomatic and functional endpoints:

- In oHCM, CMIs produce **large, rapid, reversible LVOT-gradient reductions** that track symptom and exercise gains, and benefit **reverses on drug withdrawal**

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Reversibility is the signature of a load/hemodynamic effect rather than durable substrate remodeling.

- The **discriminating prediction holds at the trial level:** in non-obstructive HCM (nHCM), where there is no gradient to relieve, the phase 3 **ODYSSEY-HCM trial did not meet its**

primary endpoint (review synthesis

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However, the "**rather than reversing the sarcomere substrate**" clause is only *partially* correct and is actively contested by three lines of evidence:

1. **The molecular target is intrinsically gradient-independent.** CMIs act on the myosin super-relaxed/off-actin (SRX) state and cross-bridge cycling in *any* myocardium

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Mechanistically there is no reason the effect should be confined to obstructed hearts.

2. **Gradient-independent biomarker signal.** In nHCM cohorts (MAVERICK-HCM, REDWOOD-HCM cohort 4) CMIs lowered cardiac biomarkers (NT-proBNP/troponin) and improved symptoms despite no gradient

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— a wall-stress/sarcomere-energetics effect, not obstruction relief.

3. **Preclinical substrate modification.** Early chronic mavacamten (MYK-461) in non-obstructed MYH7-mutant mice **prevented hypertrophy, myocyte disarray, and fibrosis**

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— direct gradient-independent substrate modification (prevention, not reversal).

The best synthesis: in **established adult oHCM**, the dominant and reproducible driver of symptomatic benefit is **obstruction relief**, and the **fibrotic substrate (LGE scar) does not clearly reverse over 6–30 months**

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But CMIs also produce **reverse hypertrophic remodeling (LV-mass regression)** and **gradient-independent biomarker effects**, and preclinical data show they can **prevent**

substrate pathology. So "chiefly obstruction relief" is accurate for *current symptomatic endpoints in obstructive disease*, but the hypothesis **overstates the dichotomy**: a genuine load-independent sarcomere/energetics component exists and is not excluded.

Most important caveats: - The decisive nHCM readout (ODYSSEY-HCM) was available only via **review-level synthesis**; the primary trial report was **not retrievable** through the search tool this run. - LV-mass regression is itself **partly load-mediated** (reduced wall stress after gradient relief), so mass change does not cleanly prove sarcomere-substrate reversal. - Several key abstracts were **truncated** (SEQUOIA CMR, EXPLORER-CN CMR, ASA-vs-mavacamten), limiting exact effect extraction.

2. Evidence Matrix

Full machine-readable table: `openscientist_artifacts/evidence_matrix.csv` (16 rows).

Summary:

PMID	Type	Stance	Mechanistic claim tested	Key finding	Subtype	Clinical
42002189	Human RCT subanalysis	Support	Load-dependent gradient relief	Aficamten –66% Valsalva LVOT-G (123→41 mmHg, p=0.001); NT-proBNP –85%; pVO2 +1.8	oHCM, very-high gradient	H w
42251960	Human RCT	Support	Benefit tracks obstruction, reversible	HCMSQ improved vs placebo (p<0.001); scores return to baseline by wk38 after stopping	oHCM (EXPLORER)	H lo
42667283	Human RCT	Support	CMI > beta-blocker functionally	Aficamten > metoprolol pVO2 (+1.9 to +3.1)	oHCM (MAPLE)	H co
41093262	Meta-analysis	Support/qualify	Class clinical effect + remodeling	4 RCTs/726: +NYHA 36%, KCCQ +8.4, pVO2 +1.6; favourable remodeling	oHCM	H fi
40624601	Review	Support + qualify	Benefit attenuated without gradient	ODYSSEY-HCM primary endpoints NOT met in nHCM; MAVERICK/ REDWOOD biomarker+symptom gains	nHCM vs oHCM	M p re
41841680	Review	Qualify/compete	CMIs don't fix energy deficit	Robust biomarker gains but no resolution of nHCM energy mismatch	nHCM	M
26912705	Model organism	Competing	Gradient-independent substrate prevention	Early chronic MYK-461 suppressed hypertrophy,	Genetic HCM mouse (non-obstructed)	H v p m

PMID	Type	Stance	Mechanistic claim tested	Key finding	Subtype	C li
				disarray, fibrosis in MYH7-mutant mice		
42715826	Meta-analysis (CMR)	Support/qualify	Structural substrate reversal	LV mass index -20.01 g/m^2 ; fibrosis impact "unclear"	oHCM (3 RCTs, n=150)	M fi
40632050	Human RCT (CMR)	Qualify	Mass + fibrosis markers	Mavacamten LV mass index -30.8 g/m^2 ; some fibrosis indicators reduced (wk30)	oHCM (EXPLORER-CN)	M tr b
41877729	Human case (serial CMR)	Support/qualify	Long-term substrate reversal	2 yr: mass down; LGE scar unchanged ; T1 down; ECV up then normalized	oHCM (n=1)	L so
39217563	Human RCT (CMR)	Support/qualify	Reverse remodeling	SEQUOIA CMR substudy (n=50): aficamten vs placebo structure/function	oHCM	M re
41688269	Human observational	Qualify	Diastolic/LA remodeling	NT-proBNP 1431→158; LAESV 50.8→39.9; LVOT 65→12	oHCM (n=55)	M g g
41590843	Human retrospective	Qualify (discriminating design)	Diastolic effect CMI vs ASA	Mavacamten vs alcohol septal ablation diastolic comparison	oHCM	M tr g in d
40000285	Review	Mechanism (competebasis)	Gradient-independent MoA	HCM = $\uparrow$ crossbridge formation + destabilized SRX; CMIs reduce crossbridges	HCM general	M

PMID	Type	Stance	Mechanistic claim tested	Key finding	Subtype	C
30371160	In vitro	Mechanism	CMI blunts hypercontractility	Myk461 reduced hypercontractile crossbridge kinetics incl cMyBPC-KO	Model myocardium	M
37060436	Review (seed)	Support (basis)	SRX is the target	Identifies SRX state in sarcomere cardiomyopathy	HCM	M

3. Data and Tool Use

Inventory of sources (see `openscientist_artifacts/search_log.md` for the full 21-query log):

- **PubMed** (NCBI E-utilities via OpenScientist `search_pubmed`). Live query, no fixed snapshot version. Retrieval date 2026-10-09. **Accessed** — abstracts retrieved for 11/21 queries; 10 queries returned no usable result; 1 HTTP 429 rate-limit. All returned PMIDs are genuinely on-topic (HCM + myosin inhibitors). Full-text not retrieved; several abstracts truncated (noted per row in `evidence_matrix.csv`).
- **ClinicalTrials.gov / trial registry API** — **NOT queried** this run. NCT IDs (EXPLORER NCT03470545, VALOR NCT04349072, SEQUOIA NCT05186818, MAPLE NCT05767346, EXPLORER-CN NCT05174416, MAVA-LTE NCT03723655) were read from abstract text, not verified against the registry. *Source absence recorded.*
- **GenCC / ClinGen / omics (GEO, GTEx, cBioPortal) / cohort datasets** — **NOT queried**. The hypothesis is a pharmacodynamic/trial-level mechanism claim, not a gene–disease validity claim, so these were out of scope. *Source absences recorded; no silent fallback occurred because no dataset analysis was attempted and abandoned.*

Analyses attempted:

Analysis	Input → Method → Output	Outcome
Literature evidence synthesis	PubMed abstracts → manual stance classification → <code>evidence_matrix.csv</code> , <code>knowledge_state</code> findings	Succeeded (literature only)
Artifact bundle authoring	findings → file authoring + <code>sha256sum</code> → <code>search_log.md</code> , <code>evidence_matrix.csv</code> , <code>environment.txt</code> , <code>MANIFEST.yaml</code>	Succeeded
Computational/statistical analysis of patient or omics data	—	Skipped (no data files provided; none available to preflight)

No statistical computation was performed; no result in this report is derived from executed analysis of primary data. All quantitative values are extracted from published abstracts.

4. Mechanistic Causal Chain

Hypothesis-implied chain (obstruction-relief dominant):

1. **Upstream trigger:** sarcomere gene mutation (MYH7, MYBPC3, ...) → destabilized myosin SRX/off-actin state → ↑ actin–myosin cross-bridges → **hypercontractility**. (*Strong:*

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2. **Hypercontractility → dynamic LVOT obstruction** (+ SAM of mitral valve). (*Strong in oHCM.*)

3. **CMI binds myosin → ↓ cross-bridge formation → ↓ contractility → ↓ LVOT gradient (load reduction)**. (*Strong, reversible:*

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4. ↓ **gradient** → ↓ **wall stress**, ↓ **filling pressure** → **symptom/exercise/biomarker improvement**. (*Strong in oHCM:*

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Where the chain is strong: steps 1–4 in obstructive disease.

Where links are inferred / missing: - **Step 3→substrate:** Does ↓ cross-bridge cycling durably **reverse** myocyte disarray / interstitial fibrosis in humans? **Inferred/absent**. Human CMR shows **mass regression** but **no clear fibrosis/LGE reversal**

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Preclinical shows **prevention** of substrate, not reversal of established disease

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- **Gradient-independent arm:** CMI → ↓ wall stress / improved energetics → biomarker/diastolic benefit **without** a gradient. **Supported in nHCM biomarkers but the clinically meaningful endpoint failed (ODYSSEY)**. This is the unresolved fork between the seed hypothesis and its competitor.

5. Knowledge Gaps

1. **Does CMI reverse the fibrotic substrate in humans?** Scope: oHCM/nHCM, longitudinal CMR (T1/ECV/LGE). Why it matters: central to the seed's "not reversing the substrate" clause. Checked: PubMed CMR queries (#9,#11,#12 null; #17,#19 informative). Current state: **mass reverses; fibrosis unclear/scar persists**. Resolver: ≥24-month RCT with ECV/T1 primary endpoint and endomyocardial sampling.
2. **ODYSSEY-HCM primary report.** Scope: the decisive nHCM readout. Why it matters: discriminates seed vs competitor. Checked: queries #2, #4, and dedicated ODYSSEY query (iter 3) — **primary report NOT retrieved via tool** (only review

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Label: **unverified absence** (tool-indexing limitation, not confirmed PubMed absence).

Resolver: retrieve the primary publication / NCT results.

3. **Gradient-independent clinical efficacy.** Scope: nHCM. Conflict: biomarker/symptom gains (MAVERICK/REDWOOD) vs failed ODYSSEY primary endpoint. Resolver: adequately powered nHCM RCT with objective endpoints; subgroup by fibrosis burden/energetic phenotype.
4. **Is LV-mass regression substrate reversal or load unloading?** Scope: oHCM. Why it matters: mass change is used as "reverse remodeling" but is confounded by wall-stress reduction. Resolver: compare CMI vs pure mechanical gradient relief (myectomy/ASA) head-to-head on mass/ECV (partially addressed by

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truncated).

5. **Genotype stratification of response.** Scope: sarcomere-positive vs -negative. Checked: query null (iter 3). If benefit were substrate-directed, genotype-positive patients might respond differently. **Data not retrieved — gap.**
6. **Source absences:** no ClinicalTrials.gov/GenCC/ClinGen/omics query this run (out of scope for a pharmacodynamic claim) — recorded, not claimed as biological absence.

6. Alternative / Competing Models

1. **Gradient-independent sarcomere-energetics model (primary competitor).** CMIs restore SRX, reduce ATP waste and wall stress, improving diastolic function and biomarkers regardless of obstruction. *Alternative* to the seed. Support:

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nHCM biomarker gains. Weakness: ODYSSEY primary-endpoint miss.

2. **Reverse-remodeling / disease-modification model.** Sustained contractility reduction → LV-mass regression, atrial reverse remodeling, possibly lower arrhythmic risk. *Downstream consequence* that may be partly load-mediated. Support:

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(SCD-risk-score effects). Weakness: fibrosis/scar unchanged.

3. **Pure hemodynamic palliation (strong seed form).** CMIs = reversible gradient relief only; stop drug → disease returns. Support: reversibility

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Weakness: cannot explain nHCM biomarker gains or preclinical substrate prevention.

4. **Metabolic/energetic substrate model (parallel mechanism).** nHCM is driven by an energy deficit CMIs do not fix; metabolic modulators (nineraxstat) target this. *Parallel/complementary*. Support:

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Implication: explains why a sarcomere-contractility drug underperforms in nHCM.

5. **Mitral-apparatus/geometry model (upstream contributor to obstruction).** AMVL length predicts CMI response

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— obstruction is partly anatomic, consistent with the seed's load-centric view in oHCM.

7. Discriminating Tests

1. **CMI vs mechanical gradient relief (myectomy/ASA), matched gradient reduction, CMR primary endpoint (ECV/T1, disarray surrogate), 24 months.** Stratify oHCM by baseline fibrosis. *Expected if seed true:* equal symptom benefit, **no** extra fibrosis/ECV improvement from CMI beyond load relief. *If competitor true:* CMI yields extra interstitial-fibrosis/energetic improvement.

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is a small retrospective start.)

2. **Adequately powered nHCM RCT (aficamten/mavacamten) with objective primary endpoint (pVO2) + CMR/biomarker secondaries,** pre-stratified by fibrosis burden and energetic phenotype (31P-MRS PCr/ATP). *Expected if seed true:* no functional benefit despite

biomarker change; *if competitor true*: functional benefit in low-fibrosis/high-energy-deficit subgroup.

3. **Genotype-stratified response analysis** (sarcomere-positive vs -negative) across pooled trials.
4. **31P-MRS myocardial energetics (PCr/ATP) before/after CMI** in oHCM and nHCM — directly tests the "sarcomere-energetics" competitor. *Expected if competitor true*: PCr/ATP improves independent of gradient.
5. **Serial endomyocardial or imaging biomarkers of myocyte disarray (diffusion-tensor CMR)** to test substrate reversal directly.

8. Curation Leads (require curator verification)

Candidate status: Keep **EMERGING**, refine label to subtype-qualified: *"In obstructive HCM, CMI symptomatic benefit is chiefly load-dependent gradient relief; a gradient-independent sarcomere/energetics effect exists but has not yet translated to a positive non-obstructive clinical endpoint."*

Candidate	evidence	references	(verify	exact	snippets):	-
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— snippet: *"the preliminary results from phase 3 trials (ODYSSEY-HCM trial) revealed that primary endpoints were not met in nonobstructive HCM"* (SUPPORT, review-level). -

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— snippet: *"In nonobstructive HCM, preliminary studies (MAVERICK-HCM trial and cohort 4 of REDWOOD-HCM trial) have reported improvements in cardiac serum biomarkers and symptoms."* (QUALIFY). -

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— snippet: *"early, chronic administration of MYK-461 suppresses the development of ventricular hypertrophy, cardiomyocyte disarray, and myocardial fibrosis"* (COMPETING). -

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— snippet: *"the impact on myocardial fibrosis remains unclear"* (SUPPORT for substrate-not-reversed). -

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— snippet: *"Therapy cessation was associated with return to baseline in HCMSQ scores by week 38."* (SUPPORT, reversibility). -

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— snippet: *"an enhanced rate of myosin-actin crossbridge formation and destabilization of the energy-conserving |"super-relaxed off-actin state|" of myosin"* (mechanism basis).

Candidate pathophysiology nodes/edges: - Node: *myosin super-relaxed (SRX) / off-actin state* → Node: *cross-bridge formation rate* → Node: *hypercontractility* → Node: *dynamic LVOT obstruction* → Node: *symptoms/wall-stress*. - Edge (well-supported): CMI → cross-bridge formation → LVOT gradient (reversible). - Edge (inferred/contested): CMI → interstitial fibrosis / myocyte disarray (preclinical-only; human-unconfirmed). - Edge (parallel): nHCM energy deficit — not addressed by CMI.

Candidate ontology terms: - Cell type: cardiomyocyte (CL:0000746); cardiac fibroblast (CL:0002548). - Processes: myosin ATPase activity (GO:0016887-related), muscle contraction (GO:0006936), cardiac muscle hypertrophy (GO:0003300), extracellular matrix / collagen fibril (fibrosis) remodeling. - Disease: hypertrophic cardiomyopathy (MONDO:0005045); subtype split obstructive vs non-obstructive.

Candidate knowledge_gaps / discussion prompts: items 1–6 in §5, especially (a) unverified ODYSSEY primary report, (b) human fibrosis-reversal unconfirmed, (c) mass-regression load-confound.

9. Limitations

- Literature-only run; no primary data analyzed. Quantities taken from abstracts, several truncated.
- The decisive nHCM endpoint relied on review synthesis (ODYSSEY primary report not retrieved via tool).
- Rapidly moving field (multiple 2026 reports); hit counts are time-sensitive (PubMed is live).

Artifact Bundle

kb/hypotheses/Hypertrophic_Cardiomyopathy/myosin_inhibitor_obstruction_vs_substrate/
openscientist_artifacts/ - MANIFEST.yaml (schema 1.0, status flags, checksums, traceability) -
search_log.md (21-query PubMed log incl. negatives) - evidence_matrix.csv (16 evidence
rows) - environment.txt (run-mode/environment record)

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